

CHAPTER 3 MOTION IN CENTRAL POTENTIAL $V(r)$

$\vec{r} = \mu r^2 \dot{\phi} \hat{\phi}$ $\vec{F} = r \hat{r}$ $\vec{p} = \mu \dot{\vec{r}}$
CONST. OF MOTION

$L = T - V = \frac{1}{2} \mu (\dot{r}^2 + r^2 \dot{\phi}^2) - V(r)$

ϕ CYCLIC $\Rightarrow \frac{\partial L}{\partial \phi} = 0 \Rightarrow p_\phi = \frac{\partial L}{\partial \dot{\phi}} = \mu r^2 \dot{\phi} = l$
CONSERVED

$V_{\text{eff}}(r) = V(r) + \frac{l^2}{2\mu r^2}$

$E = T + V = \frac{1}{2} \mu (\dot{r}^2 + r^2 \dot{\phi}^2) + V(r) = \frac{1}{2} \mu \dot{r}^2 + V_{\text{eff}}(r)$

UNIVERSAL METHOD $\Rightarrow$ COMPLETE SOLUTION

$t = \int_0^r \frac{dr}{\sqrt{\frac{2}{\mu} (E - V_{\text{eff}}(r))}}$ $t = t(r)$

$\phi = \phi_0 + \int_r^r \frac{l}{\mu r^2 \dot{r}} dr$ $\phi = \phi(t)$

LHS: $\mu \ddot{r} = -\frac{l^2}{\mu} \frac{1}{r^3} \frac{du}{d\phi} = -\frac{l^2}{\mu} u^2 \frac{d^2 u}{d\phi^2}$

RHS: $F(r) + \frac{l^2}{\mu r^3}$

LHS=RHS: $-\frac{l^2}{\mu} u^2 \frac{d^2 u}{d\phi^2} = F(1/u) + \frac{l^2}{\mu} u^3$

GENERIC EQ. OF TRAJECTORY FOR ARBITRARY $V(r)$

DIFF EQN FOR $r = r(\phi)$

$l = \mu r^2 \dot{\phi} \Rightarrow \dot{\phi} = \frac{l}{\mu r^2} = \frac{d\phi}{dt}$

$\frac{dr}{dt} = \frac{dr}{d\phi} \frac{d\phi}{dt} = \frac{l}{\mu r^2} \frac{dr}{d\phi}$

$\frac{dr}{dt} = -\frac{l}{r^2} = -u^2$

$F(r) = F(1/u) = -\frac{dV}{dr} = u^2 \frac{dV(1/u)}{du}$

$-F(1/u) = -u^2 \frac{dV(1/u)}{du}$

$-F(1/u) = \frac{l^2}{\mu} u^2 \left[u + \frac{du}{d\phi^2} \right]$

$-u^2 \frac{dV}{du} = \frac{l^2}{\mu} u^2 \left[u + \frac{du}{d\phi^2} \right]$

$\frac{d^2 u}{d\phi^2} + u = -\frac{\mu}{l^2} V(1/u)$

EQ FOR TRAJECTORY $r(\phi)$

SOLVE FOR GIVEN $u = u_0$ FOR $\phi = \phi_0$ $\frac{du}{d\phi} = 0$

$\frac{du}{d\phi} = 0 \Leftrightarrow$ MAX OR MIN r POINT

ABSOLUTE POINTS

$r = \text{MAX OR MIN}$

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$F(r) = -\frac{\partial V}{\partial r}$

$\frac{dV}{dr} = \frac{dV}{du} \frac{du}{dr} = -u^2 \frac{dV}{du}$

$\frac{du}{dr} = -\frac{1}{r^2} = -u^2$

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GRAVITATION CASE KEPLER PROBLEM

$V = -\frac{k}{r} = -ku$ $\frac{dV}{du} = -k$

$\frac{d^2 u}{d\phi^2} + u = \frac{k\mu}{l^2}$

HARMONIC OSCILLATOR WITH SHIFTED ORIGIN

ASIDE: $u = u - \frac{k\mu}{l^2}$

HOMOGENEOUS SOLUTION: $\tilde{u}(\phi) = A \cos(\phi - \phi_0)$

$\tilde{u}(\phi) = -A \sin(\phi - \phi_0)$

$\tilde{u}(\phi) = -A \cos(\phi - \phi_0)$

$\tilde{u}(\phi) = -\tilde{u}(\phi)$

$u(\phi) = \frac{k\mu}{l^2} + A \cos(\phi - \phi_0)$

$u_0 = \frac{1}{r_0}$

$\frac{du}{d\phi} = 0$

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$u(\phi) = \frac{k\mu}{l^2} [e \cos(\phi - \phi_0) + 1] = \frac{1}{r}$

e CONSTANT $e \equiv \frac{A}{k\mu/l^2}$

USING THE UNIVERSAL METHOD

$u'' = -u + \frac{k\mu}{l^2}$

$\nabla \times u(\phi)$

$\frac{du}{d\phi} = \sqrt{\dots}$

$\frac{du}{d\phi}$

$u = \frac{k\mu}{l^2} [1 + \sqrt{1 + \frac{2l^2 E}{\mu k^2}} \cos(\phi - \phi_0)] = \frac{1}{r}$

$e \equiv \sqrt{1 + \frac{2l^2 E}{\mu k^2}}$